Nonlinear element

W21 (2021) — English translation

The electrical circuit consists of a constant voltage source U_0, a capacitor of capacitance C, a nonlinear element NE, and a switch K. The current-voltage characteristic of the nonlinear element is as follows:

$$\left(\frac{I}{I_0}\right)^2 + \left(\frac{U}{U_0}\right)^2 = 1$$

for $I > 0$ and $I = 0$ for $|U| \geq U_0$ (see figure). The value I_0 is known. At the initial moment, the voltage across the capacitor is $U_0/2$. The switch K is closed.

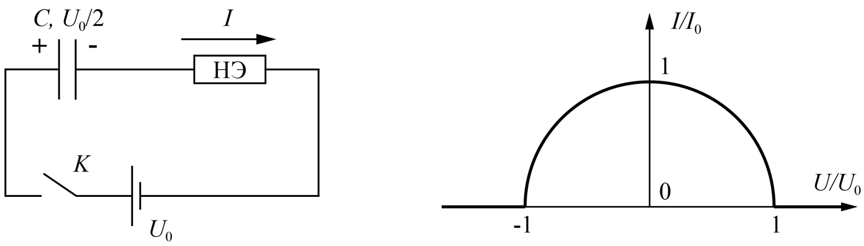

Figure 1: Figure 1.

A1	Find the maximum power P_1 released on the nonlinear element.	**2.00**

A2	Find the maximum rate of change of energy of the capacitor P_2 and the time after closing the switch Δt after which it is achieved.	**8.00**

Source: `https://pho.rs/p/80`